Derivatives

Gradient

$$\operatorname{grad} f = \nabla f = [f_x, f_y, f_z]$$

Directional Derivative

$$D_{\mathbf{a}} f = \frac{df}{ds} = \frac{\mathbf{a}}{|\mathbf{a}|} \nabla f$$

Divergence

$$\operatorname{div} \mathbf{v} = \nabla \cdot \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$\nabla \cdot \mathbf{v} = 0 \rightarrow$ **incompressible**

Curl

$$\operatorname{curl} \mathbf{v} = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Laplacian

$$\Delta f = \nabla \cdot (\nabla f) = \nabla^2 f = f_{xx} + f_{yy} + f_{zz}$$

Common Equalities

$\nabla fg = f \nabla g + g \nabla f$

$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$

$\nabla \cdot (f\mathbf{v}) = f \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla f$

$\nabla \cdot (f \nabla g) = f \Delta g + \nabla f \cdot \nabla g$

$\Delta(fg) = f \Delta g + 2 \nabla f \cdot \nabla g + g \Delta f$

$\nabla \times (f\mathbf{v}) = \nabla f \times \mathbf{v} + f(\nabla \times \mathbf{v})$

$\nabla \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\nabla \times \mathbf{u}) - \mathbf{u} \cdot (\nabla \times \mathbf{v})$

$\nabla \times (\nabla f) = \mathbf{0}$

$\nabla \cdot (\nabla \times \mathbf{v}) = 0$

Normal Vector

Case $f(x, y, z) =$ **const:** $\mathbf{N} = \nabla f$

Case $z = f(x, y)$**:** $\mathbf{N} = [f_x, f_y, -1]$

Tangent Plane

At point (x_0, y_0, z_0), the tangent plane is

$$N_x(x - x_0) + N_y(y - y_0) + N_z(z - z_0) = 0$$

Line Integrals

Arc Length $ds = |\mathbf{r}'(t)|\ dt$

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\ dt$$

$$\int_C (F_1\ dx + F_2\ dy + F_3\ dz) = \int_a^b (F_1 x' + F_2 y' + F_3 z')\ dt$$

The line integral is **path independent** iff,

1. Field is **conservative**: $\mathbf{F} = \nabla f$
2. Value around all closed paths is **zero**
3. Field is **exact**: $\nabla \times \mathbf{F} = \mathbf{0}$

Note: If $\mathbf{F}$ is defined on a *simply connected domain* (i.e. no holes), then $\mathbf{F}$ is conservative $\iff$ $\mathbf{F}$ is exact

If path independent,

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_A^B \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

Finding f such that $\mathbf{F} = \nabla f$

1. Verify the field is exact.
2. Integrate one component of $\mathbf{F}$ wrt the corresponding variable. The result is a fn plus an unknown fn of the other variables. Ex.
$$f = \int F_1 dx + g(y, z)$$
3. Take the partial of the next variable and equate to corresponding component.
4. Rinse & repeat. If functions turn out to be a constant, choose zero.

Green's Theorem

$$\iint_R \nabla \times \mathbf{F} \cdot \mathbf{k}\ dA = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)\ dx\ dy = \oint_C (F_1\ dx + F_2\ dy)$$

Area Formulas

Area $= \oint_C x\ dy$

Area $= -\oint_C y\ dx$

Area $= \frac{1}{2} \oint_C (x\ dy - y\ dx)$

Area $= \frac{1}{2} \oint_C r^2\ d\theta$

Transforming the Laplacian

$$\iint_R \Delta w\ dA = \oint_C \nabla w \cdot \mathbf{n}\ ds = \oint_C \frac{\partial w}{\partial n}\ ds$$

Surfaces

$$\mathbf{r} = \mathbf{r}(\mathbf{u}, \mathbf{v})$$

$$\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$$

Cylinder $x^2 + y^2 = a^2, \quad |z| \le 1$

$$\mathbf{r}(z, \theta) = \begin{bmatrix} a\cos\theta \\ a\sin\theta \\ z \end{bmatrix}, \quad \theta \in [0, 2\pi]$$

Sphere $x^2 + y^2 + z^2 = a^2$

$$\mathbf{r}(\phi, \theta) = \begin{bmatrix} a\sin\phi\cos\theta \\ a\sin\phi\sin\theta \\ a\cos\phi \end{bmatrix}, \quad \begin{matrix} \phi \in [0, \pi] \\ \theta \in [0, 2\pi] \end{matrix}$$

Cone $z = \sqrt{x^2 + y^2}$

$$\mathbf{r}(r, \theta) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \\ r \end{bmatrix}, \qquad \begin{matrix} r \in [0, H] \\ \theta \in [0, 2\pi] \end{matrix}$$

Torus

With major radius R, and minor radius r

$$\mathbf{r}(\phi, \theta) = \begin{bmatrix} (R + r\cos\phi)\cos\theta \\ (R + r\cos\phi)\sin\theta \\ r\sin\phi \end{bmatrix}, \qquad \begin{matrix} \phi \in [0, 2\pi] \\ \theta \in [0, 2\pi] \end{matrix}$$

Surface Integrals

Area element $dA = ||\mathbf{r}_u \times \mathbf{r}_v||\ du\ dv = ||\mathbf{N}||\ du\ dv$

Flux

The amount of fluid passing through a surface S per unit time. S can also be parameterized as $\mathbf{r}(u, v)$ for $(u, v) \in \Omega$.

$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n}\ dA = \iint_\Omega \mathbf{F}(\mathbf{r}) \cdot ||\mathbf{N}||\ du\ dv$$

Surface Area

$$\text{Surface Area} = \iint_\Omega ||\mathbf{N}||\ du\ dv$$

Scalar Surface Integral

$$\iint_S G\ dA = \iint_\Omega G(\mathbf{r})||\mathbf{N}||\ du\ dv$$

Divergence Theorem

Let V be a solid region with boundary S oriented outward. If $\mathbf{F}$ is continuously differentiable,

$$\iiint_V \nabla \cdot \mathbf{F}\ dV = \iint_S \mathbf{F} \cdot \mathbf{n}\ dA$$

Stokes' Theorem

Let S be an oriented surface with right-hand boundary C. If $\mathbf{F}$ is continuously differentiable,

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n}\ dA = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Triple Integrals

Rectangular: $dV = dx\ dy\ dz$

Cylindrical: $dV = r\ dr\ d\theta\ dz$

Spherical: $dV = \rho^2 \sin\phi\ d\rho\ d\theta\ d\phi$

Green's Identities

Derived by applying the Divergence Theorem to the vector field $\mathbf{F} = f\,\nabla g$.

Green's First Identity

$$\iiint_T (f\,\Delta g + \nabla f \cdot \nabla g)\ dV = \iint_S f \frac{\partial g}{\partial n}\ dA$$

Green's Second Identity

$$\iiint_T (f\,\Delta g - g\,\Delta f)\ dV = \iint_S \left(f\frac{\partial g}{\partial n} - g\frac{\partial f}{\partial n} \right)\ dA$$